



FRD-LSTM: a novel technique for fake reviews detection using DCWR with the Bi-LSTM method

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Abstract

The growth of the internet and the availability of online stores has increased the trend of online shopping among customers. People preferred to read the comments posted by potential buyers before purchasing a product. However, online stores are using machine learning (ML)-based approaches to generate fake comments about the product with the intention of either creating a false positive impact about their products or hitting the reputations of competitors by posting negative comments about their products. Therefore, it is crucial to classify real and fake reviews from online stores to save customers from fraud. To overcome these challenges, we have presented a deep learning-based approach namely FRD-LSTM to timely recognize fake reviews. After performing the preprocessing step, the deep contextualized word representation (DCWR) approach is applied to compute the deep features. Then, the principal component analysis (PCA) approach is applied to minimize the feature space. Finally, the Bi-LSTM classifier is trained on the computed features to classify the real and bogus reviews. Our method improves the fake reviews classification performance while decreasing both the training and testing time complexity. We have tested our approach on a challenging dataset namely the Amazon product reviews database and attained an average accuracy value of 97.21%. We have confirmed through extensive experimentation that our approach is proficient to detect bogus reviews and can assist online buyers in protecting them from fake products.

Keywords Fake reviews · Deep learning · PCA · Bi-LSTM

1 Introduction

Nowadays online shopping become a trend in the world. People usually buy things online and these online shopping sites have review options for customer feedback. People usually decide

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to buy or not purchase products on basis of already available reviews from customers. According to some recent studies [20], it is reported that about 81% of people read and trust the reviews of products before purchasing them. Because of the such prevailed impact of reviews, several online vendors intend to post fake positive reviews about their products while posting fake negative reviews for opponents' products to obtain financial benefits. Fake or bogus reviews are generated by using two methods which are as follows: In the first method, humans are hired to write scripted reviews about a product. In such scenarios, the appointed reviewers may have never seen the original product but still, give paid reviews about it. While in the second approach, some computer-based automated text generation approaches are used to automatically write fake reviews about the products [21]. Conventionally, human-written paid reviews can be written in a limited amount and the company owners can hire spammers in limited numbers for posting fake reviews. However, the great advancement in the area of machine learning and natural language processing (NLP) has introduced such remarkable algorithms which can post unlimited scripted reviews about any product at a fast pace in comparison to humans [36].

The such false generated information about the products can affect digital marketing and e-commerce in three main ways. Firstly, such bogus reviews may affect the trust of buyers which could result in a severe market failure. Genuine buyers post authentic reviews to communicate their shopping experience which can be positive or negative depending on their level of satisfaction with products [18]. Such reviews bring a valued service to both the marketplace and sellers to assess the quality of their products. Whereas scripted reviews can damage the trustworthiness of buyers like someone who wants to buy a laptop and the reviews are very positive but actually, the laptop is damaged. Such paid product promotions can systematically demean source authenticity and cause the wrong selection of products which in turn causes the loss of both buyer money and trust in the purchased product and its brand. Secondly, such scripted reviews can have an impact on the ranking of products either in a positive manner (if the generated reviews are positive) or in a negative manner (if the bogus reviews are negative) [2]. This is due to the reason that the algorithms employed to rank the online markets utilize these reviews as a measure to estimate the ranking of a product against its competitors. Hence, such generated false content may cause biased competition, in which the ranking of a product is falsely overstated or collapsed. So, such strategies can be used as a weapon by an organization to affect the reputation of its rivals. Lastly, the fake reviews will not only impose a superficial or reputational impact but accompanies a monetary cost as well. Like, a study conducted in [29] shows that a one-star reduction in a brand's Yelp ranking shows a five to 9 % decline in income. Hence, the generation of false reviews is imposing a serious threat to both companies' profit and buyers' well-being. Therefore, automatic spam identification is an essential task because it can protect buyers from a waste of money and time. Even though several studies have been conducted to identify the fake generated reviews, however, this type of spam identification is difficult because spammers can easily hide their identities [19, 25]. So, there is a need for such a method that can timely detect or identify the spammers of false reviews to save the buyers money [15].

The researchers are continually analyzing the online posted comments to check them for realism and fake to assist the users in better understating the quality of various products. Meanwhile, the great progression in AI has insisted the research community use DL-based approaches for classifying real and bogus reviews of several products. However, the classification performance needs further improvements due to the improved realism of posted comments which can easily ditch the consumers. This work introduces an effective DL

approach named FRD-LSTM for identifying real and fake product reviews. More clearly, we have used the DCWR approach with the PCA algorithm for text features computation which is later classified by the Bi-LSTM classifier. The main contributions of the presented work are listed as follows:

- Introduced a DL-based technique named FRD-LSTM in which DCWR for feature calculation, improves the fake reviews classification performance while decreasing both the training and testing time complexity.
- An efficient approach to fake reviews identification using the PCA algorithm to minimize the feature vector space by nominating a reliable set of textual features.
- Obtained better classification performance due to the usage of deep features with the Bi-LSTM approach that responsibly represents the complicated text transformations.
- Effective classification accuracy is due to the better features nomination power of the FRD-LSTM model which enables it to better deal with the over-fitted model training data.
- Extensive experiments have been carried out as compared to other latest fake reviews classification approaches by using a standard database containing diverse and unbalanced data to demonstrate the accuracy of the introduced model.

2 Related work

Many researchers have attempted to propose methods for the timely detection of fake reviews. In this section, we have performed an analysis of existing techniques used for fake reviews detection and classification. One such method was proposed in [32], where initially, a preprocessing step was performed to clean the data. Then the three-dimensional time-series approach was used to compute the text and behavioral features of online reviews. Finally, the random forest (RF) classifier was trained on the calculated features to classify the real and fake reviews. The approach shows better fake reviews classification performance, however, the technique requires more evaluations to show its robustness. Salminen et al. [26] proposed a method for fake review recognition. A model namely the OpenAI fake detection approach along with the SVM classifier was used for identifying the bogus reviews. The approach [26] works well with machine-generated fake reviews, however, unable to perform well in detecting human-generated fake reviews. Adelani et al. [1] presented a sentiment-observable bogus reviews creation and detection approach. Initially, the GPT-2 framework was applied to produce extensive reviews with the preferred sentiment chosen from the target website. In the next step, the BERT text classifier was applied to remove the unwanted sentiments. After this, three approaches namely Grover, GLTR, and GPT-2 classifiers were nominated for bogus review identification. The proposed approach [1] shows an equal error rate of 22.5% for fake reviews detection which requires improvements to better tackle complex real-world scenarios. Another approach for the timely identification of online generated fake reviews was presented in [12] where the concept of ensemble approaches was used. More specifically, the bagging and boosting approaches were combined with the SVM classifier to detect and classify fake reviews. The approach [12] is robust to detect fake reviews from the static text, however, unable to locate the bogus generated data in an interactive session among different parties. Another approach was proposed in [8] to detect the fake generated reviews from the websites. In the first step, a preprocessing step was applied to remove the unnecessary details from the input data. Then, the textural features were extracted from the processed data. After this,

several ML-based classifiers namely the KNN, Naive Bayes (NB), RF and the SVM classifiers were trained on the computed keypoints to accomplish the classification task. The approach [8] obtains the best results for the KNN classifier, however, lacks to generalize well to real-world scenarios.

Gupta et al. [11] proposed a technique to locate the bogus reviews from the original reviews by using ML-based approaches. The method employed several pre-trained frameworks namely the BERT, RoBERTa, ALBERT, and DistilBERT to classify the real and bogus reviews. The approach [11] acquires the best results for the RoBERTa model, however, the classification results need more improvements. Alsubari et al. [5] introduced a CNN-LSTM framework to efficiently locate the real and fake reviews from the websites. Initially, a CNN model was proposed to compute the features and minimize the data dimensions as well. Then, the LSTM architecture was introduced and joined with the CNN model to extract the contextual information of n-gram keypoints. Lastly, a classification layer was added to categorize the original and bogus reviews. The work presented in [5] performed well for the fake reviews identification, however, requires a huge amount of training data which ultimately enhances the computational cost. Alsharif et al. [4] presented a method to identify the fake reviews written on e-commerce websites to mislead consumers. The work initially performed the preprocessing step to remove the punctuation marks and contractions from the employed dataset. The linguistic inquiry and word count dictionary were employed to compute extra-linguistic keypoints from the dataset. The extracted features were passed to the LSTM module to categorize the real and fake reviews. The approach [4] improves the fake reviews classification performance, however, requires a huge amount of training data. Another technique to locate the fake reviews was proposed in [35] where the ensembling of several classifiers was used to perform the classification task. The method [35] comprised four main steps where first a data resampling step via using the grid search approach was performed to tackle the issue of data unbalancing. Secondly, the key points pruning was performed to eliminate the least important features. Then the final feature set was computed after applying the Parameters optimization approach. Finally, the extracted features were used to train the several classifiers namely RF, SVM, CNN, and LSTM, and the final decision was made on the majority voting strategy. The approach [35] works well for the fake reviews classification, however, suffers from high computational cost.

Another method for bogus review identification was presented in [31], in which the computed features were passed to numerous classifiers namely the SVM, KNN, RF, DT, and LDA to categorize the original and generated reviews. The approach [31] shows the best results for the SVM classifier, however, may not generalize well to unseen cases. Several techniques have been reported by the researchers for the correct identification of bogus reviews, however, several approaches lack to integrate the link between semantics and time of the given reviews. Moreover, the influence of several sources of information for bogus reviews was not taken into count. Such information establishes a complicated, extensive, diverse relation among assessors, reviews, stores, and supplies. To tackle such issues a work was presented in [9] where a dynamic knowledge graph-based approach for bogus reviews detection. The model [9] computed four kinds of entities employing a developed NN framework namely the sentence vector/twin-word embedding-based LSTM model. The model [9] works well for the fake reviews detection, however, at the charge of the enhanced computational burden. Another approach was proposed in [28] where a multi-channel deep learning model namely Mc-DNN, was proposed for classifying fake or real news, however, at the expense of increased economic cost. Javed et al. [22] introduced a CNN-based approach for

the efficient identification of fake reviews given on online stores to mislead consumers. The n-gram embeddings were passed as input to the CNN model to compute the deep features and perform the classification task. The technique presented in [22] is robust to fake reviews detection, however, unable to accurately locate human-written bogus reviews. Li et al. [18] presented a review clustering approach to efficiently divide the reviews posted by the reviewers' into groups to correctly locate the real and fake reviews. The technique [18] works well for fake reviews detection, however, classification performance needs improvements. Similarly, Tufail et al. [29] presented a framework to analyze the impact of bogus reviews on e-commerce. Initially, data preprocessing was performed to remove unnecessary details from the employed database. Then, the keypoints were nominated based on the length count, Bigram Type and sentiment-based word counts etc. Finally, several classifiers namely KNN, SVM and logistic regression (LR) were trained on the calculated features to perform the classification job. The method [29] attains the best results for the SVM classifier, however, it requires the evaluation on a more challenging dataset.

Although extensive work has been presented to predict fake reviews via employing user reviews. However, these methods either use a small amount of data for model evaluation. Moreover, these studies require improvements in classification performance in tackling real-world problems. To deal with the aforementioned problems, there is a need for such a technique that can provide an efficient and effective solution to fake reviews prediction. In this work, we have tried to cover these gaps and presented an approach that has enhanced the fake reviews prediction performance and has evaluated over a large and diverse set of examples.

3 Proposed method

In this work, we have presented a deep learning-based approach for fake review detection and classification. This method (named FRD-LSTM) is based on three main modules, i.e., Preprocessing of data, feature extraction using the DCWR method along with feature reduction approach, and classification of fake reviews is performed using the LSTM classifier. The complete flow of our methodology is shown in Fig. 1.

3.1 Preprocessing

Mostly the data has different unwanted phrases or words like misspelling, stop words, etc., which can affect the performance of the technique and increases the computation cost. To solve this problem, we have performed preprocessing steps which include stop-word removal, spelling correction, tokenization, and noise removal [16]. Our preprocessing involves the following important steps:

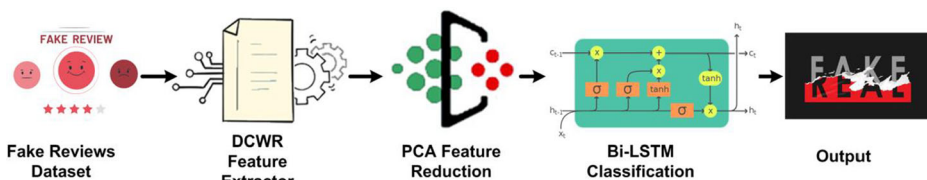


Fig. 1 Flow diagram of proposed method

- The common stop words in textual data are: “the, and, for, a, about, after”, and do not have any vital impact on classification approaches. We have removed these types of words from our data which can help in the reduction of features in a properly sized technique.
- The special symbols and punctuations are considered noise in textual data analysis, which can affect the classification task. So, we have cleaned these types of noisy data from our original dataset.
- The lowercase format is essential for text analysis because it can provide accurate comparative procedures and saves execution time. For this purpose, we have converted the whole textual data into lowercase.
- To convert or break the text into words, phrases, and tokens is called Tokenization. The purpose of this task is to break down the text into adaptable pieces for an efficient training procedure.
- The text data has different types of infections in sentences like happily and happiness. These words are handled by reducing them from root (like happi) through the technique which utilizes the simple heuristic approach. The root can be a canonical form of the original word. This process is called “Stemming” which can improve the efficiency of search methods.
- Lemmatization is the process of stemming from the original root form. It will map the words to their root form like happy, which should be Grammatically correct.

3.2 Features extraction using DCWR

After preprocessing, we extracted the important features using deep contextualized word representation (DCWR) [16, 27]. Our model DCWR is based on two-layered bidirectional LSTM and computation of word vector. Embeddings from Language Models (ELMo) is utilized for the computation of word vector which contains attributes of word usage and their deployment methods. The primary purpose of the usage of ELMo is that the resultant vectors are calculated through the bidirectional language model (biLM), which contains the backward and forward language models [3].

The forward LM can be estimated as follows:

$$p(t_1, t_2, t_3, \dots, t_N) = \prod_{k=1}^N p(t_k | t_1, t_2, t_3, \dots, t_{k-1}) \quad (1)$$

And the backward LM is calculated through the following equation:

$$p(t_1, t_2, t_3, \dots, t_N) = \prod_{k=1}^N p(t_k | t_{k+1}, t_{k+2}, t_{k+3}, \dots, t_N) \quad (2)$$

From Eqs. 1 and 2, we get LMs simultaneously which develop the log-likelihood in both directions:

$$\sum_{k=1}^N \left(\log p(t_k | t_1, \dots, t_{k-1}; \Theta_x, \vec{\Theta}_{LSTM}, \Theta_s) + \log p(t_k | t_{k+1}, \dots, t_N; \Theta_x, \Theta_{LSTM}, \Theta_s) \right) \quad (3)$$

Here token representation is Θ_x , while the softmax layer is represented by Θ_s .

Now, the job-specific weights after all the biLM layers are applied to find the ELMo as in Eq. 4:

$$ELMo_k^{task} = E(R_k; \Theta^{task}) = \gamma^{tas} \sum_{i=0}^L S_j^{task} h_{k,j}^{LM} \quad (4)$$

S^{task} is the softmax-normalized weight and γ^{task} is a scaling parameter.

Eq. 5 presented the calculation of h .

$$h_{k,j}^{LM} = \left[\overrightarrow{h}_{k,j}^{LM}, h_{k,j}^{LM} \right] \quad (5)$$

3.3 Feature reduction

After performing the key points calculation step, a feature minimization step is performed as the computed feature vector is facing the issue of large dimensional space which in turn imposes a huge computational burden for the text categorization task. To perform this task, we have used the PCA approach as originally it is employed for signal processing, but it has shown tremendous performance in analyzing the textual content as well [24]. The PCA algorithm is a well-known technique of linear algebraic application that is employed extensively in all analytical forms. The basic reason to choose the PCA for our task is its simple architecture and nonparametric behavior that results in selecting the more nominative features from the high dimensional space. The basic idea of PCA is to examine the major components of minimizing the features and locating a communication link between the selected features. The nominated components contain the highest variance in the feature space and are independent of each other. The mathematical form of PCA is given as:

$$\begin{cases} X_1 = P_{11}v_1 + P_{12}v_2 + \dots + P_{1k}v_k \\ X_2 = P_{21}v_1 + P_{22}v_2 + \dots + P_{2k}v_k \\ \dots \\ X_m = P_{m1}v_1 + P_{m2}v_2 + \dots + P_{mk}v_k \end{cases} \quad \text{Where } m \leq k \quad (6)$$

Here, $X_1, X_2, \dots, X_m$ are presenting the newly computed key points, while m is showing the total number of features. $v_1, v_2, \dots, v_k$ are the original features where the k is showing the total number of original features.

3.4 Classification

After extracting the required features, the next step is to perform the classification task to categorize the real and fake customer reviews. For this reason, we have used the Bi-LSTM approach which is an extended form of the RNN DL method [6]. The LSTM model consists of different deep layers comprising units and gate cells for storing and flowing the information in memory cells. The traditional RNN model is capable of simplifying consecutive data processing and is mainly used in the area of speech and text analysis. However, the main limitation of the conventional RNN model is its inability to perceive long-term dependencies. To deal with

the problems of the RNN approach, the LSTM model is presented which is capable of processing variable-length sequences of data like customer reviews. The LSTM approach employs integrated units to save the gained information to be used to perform future classification tasks [30].

In our work, we have used the Bi-LSTM framework that worked by joining the outputs of two different RNN layers. In Bi-LSTM, one layer examines the data sequence from right to left while the other process it from left to right. The Bi-LSTM model comprises three main modules which are the input, output, and forget gates respectively. A generic representation of the Bi-LSTM cell is given in Fig. 2.

The major reason to select the Bi-LSTM approach for our work is that it is empowered to process the long sequences efficiently by considering the information in both directions. To implement the Bi-LSTM approach, the embedding and dense layers are added which are provided by Keras, an open-source python-based library. After adding the Bi-LSTM layer, an embedding layer is added that generated a fixed-length feature vector for all data sequences. Moreover, a dropout layer is introduced to avoid the model overfitting problem. Furthermore, the sigmoid activation function is used with the dense layer to generate binary output (i.e., real or fake reviews). The detailed architecture of Bi-LSTM is shown in Fig. 3.

4 Evaluation results

In this section, we have elaborated on the details of the dataset used to test the proposed framework. We have also reported the evaluation measures used for assessing the fake reviews detection power of our work. Furthermore, a detailed comparative performance analysis is conducted to show the robustness of the proposed approach.

4.1 Dataset

To evaluate the proposed model, we have used the Amazon product reviews dataset which contains a total of 21,000 reviews [14]. This database contains 10,500 real and 10,500 fake or bogus reviews for several categories of products. All reviews of this dataset contain product Id, product and reviewer names, product ratings, purchase status as being verified or not, and associated class labels. Statistically, the Amazon product reviews dataset contains an average

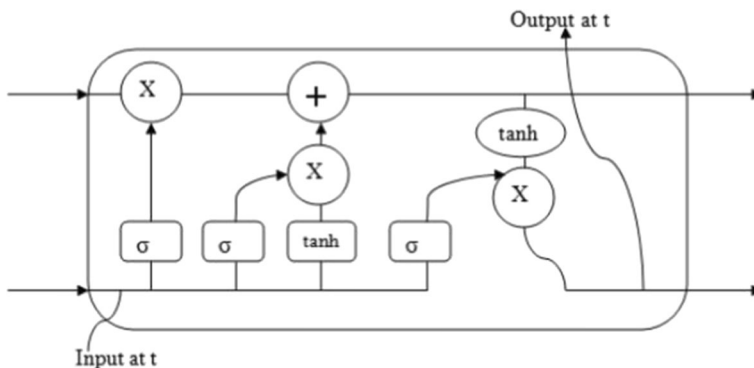


Fig. 2 Architectural view of Bi-LSTM cell [6]

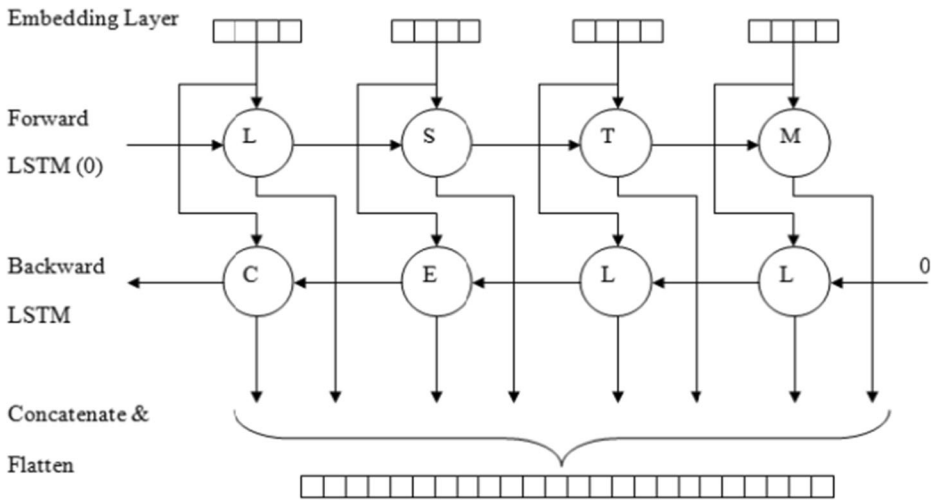


Fig. 3 Visual depiction of the Bi-LSTM structure [6]

product rating of 4.13, while about 55.70% of the product purchased are recognized as verified. The dataset contains balanced reviews for all types of products as each product comprises 700 reviews with 350 as real and 350 as bogus. Moreover, the Amazon filtering algorithm is used for labeling this database that is utilized by the Amazon website.

4.2 Performance measures

To validate the fake reviews classification performance of the proposed solution, several evaluations are conducted on the complex database named Amazon product reviews dataset by using the standard measures namely Precision (P), Recall (R), F1-Measure, and accuracy (A). The mathematical formulation of all used measures is explained in Equations from (7) to (10) respectively.

$$P = \frac{Tp}{Tp + Fp} \quad (7)$$

$$R = \frac{Tp}{Tp + Fn} \quad (8)$$

$$F1\text{-Measure} = \frac{2 \times P \times R}{P + R} \quad (9)$$

$$A = \frac{Tp + Tn}{Tp + Fp + Tn + Fn} \quad (10)$$

Here, Tp , Fp , Tn , and Fn show the true and false positive, and true and false negative respectively.

4.3 Results

In this section, we have performed several experiments to explain the fake review recognition performance of the presented work.

4.3.1 Evaluation of the proposed model

For an accurate reviews classification system, it must be competent enough to identify real and bogus user reviews. To check this, we have tested our approach over the Amazon product reviews dataset to determine its fake reviews recognition power by computing several standard measures.

Initially, we calculated the precision, recall, and accuracy of our approach for the fake reviews detection, and obtained results are shown with the help of the box plots. As the box plots are well-organized to deeply describe the results by showing the maximum, minimum, and average values of the performance parameters (Fig. 4). The values presented in Fig. 4 are clearly showing that our method is proficient in categorizing the real and bogus reviews. Furthermore, we have reported the class-wise F1-Score in Fig. 5 along with the error rate. It is quite evident that the proposed work is robust to fake reviews classification and exhibits the minimum error rate of 2.90%.

Additionally, we have drawn the confusion matrix to further demonstrate the review classification results of our approach as being real or fake, as the confusion matrix can better show the classification performance by exhibiting the original and predicted results (Fig. 6). More descriptively, we have obtained the average TPR rates of 98.13%, and 96.36% for the real and fake reviews detection respectively. From all the conducted analyses, it can be concluded that our work can accurately differentiate the real and fake reviews due to the employment of efficient feature extraction with the DCWR feature extractor and the better classification ability of the Bi-LSTM classifier.

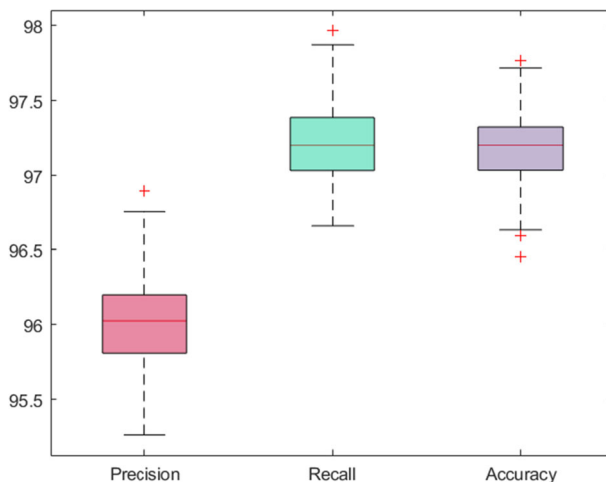


Fig. 4 Obtained results for the FRD-LSTM model

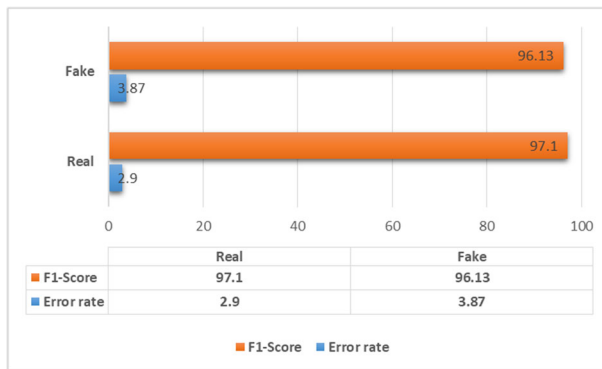


Fig. 5 F1-Score along with the error rates for the proposed approach

4.3.2 Comparative analysis with ML-based classifiers

We experimented to compare the performance of our approach in classifying real and fake user reviews with several ML-based classifiers. For this reason, numerous ML classifiers namely the SVM [22], Logistic Regression [34], KNN [10], Naïve Bayes [33], and Random Forest [23] are taken for comparison as stated in [17]. The results are compared with the help of standard evaluation measures namely the precision, recall, and F1-Score and obtained results are presented in Table 1. It is quite visible from Table 1 that our approach outperforms the other classifiers in terms of all performance metrics with the precision, recall, and F1-Score values of 96%, 97.24%, and 96.61% respectively. The second highest results are obtained by the logistic regression classifier with the precision, recall, and F1-Score values of 72.20%, 70.10%, and 72.30% respectively. While the lowest results are acquired by the Random Forest classifier with the precision, recall, and F1-Score values of 65.50%, 67%, and 69% respectively. More clearly, for the precision metric, the comparative approaches show an average value of 67.58% which is 96% for our approach. Hence, for the precision evaluation measure, we have given a 28.42% performance gain. Similarly, for the recall and F1-Score, the competitor algorithms demonstrate the average values of 67.82%, and 71.60% respectively which are 97.24%, and 96.61% for the proposed solution. Therefore, we have shown a 29.42%, and 25.01% performance gain for the recall, and F1-Score respectively. The results are clearly showing that our model is more accurate to fake reviews classification due to its power to

Fig. 6 Confusion matrix

True Class	Real	98.13%	3.64%
	Fake	1.87%	96.36%
		Real	Fake
		Predicted Class	

Table 1 Comparison of the FRD-LSTM model with ML-based classifiers

Classifier	Precision	Recall	F1-Score
SVM	67.80%	69.90%	72.70%
Logistic Regression	72.20%	70.10%	72.30%
KNN	64.20%	71.40%	72.10%
Naïve Bayes	68.20%	60.70%	71.90%
Random Forest	65.50%	67%	69%
Proposed (FRD-LSTM)	96%	97.24%	96.61%

nominate the representative set of text features and employment of the Bi-LSTM classifier which enables it to better deal with the text transformations.

4.3.3 Comparative analysis with the latest approaches

In this section, we have performed a comparison to evaluate the proposed framework against several state-of-the-art methods from history presented for fake reviews classification that employed the Amazon product reviews dataset. For a fair comparison, the average results of the proposed approach are compared with the selected studies discussed in [5, 7, 13, 17]. The comparative analysis is shown in Table 2 with the help of numerous standard performance measures. Alsubari et al. [5] proposed a CNN-LSTM approach for classifying the fake and real reviews from online shopping websites and attained an average accuracy value of 87% along with precision, recall, and F1-Score of 87%, 85%, and 86% respectively. Similarly, the work in [13] employed a deep learning approach by combining word embedding and emotion mining to classify the original and bogus product reviews. The technique [13] attained an average accuracy value of 89.30% along with the F1-Score of 82.50%. Kumar et al. [17] proposed a probabilistic approach for bogus reviews recognition and showed an average accuracy value of 78.20% along with precision, recall, and F1-Score of 73%, 73.80%, and 73.30% respectively. While the approach in [7] used a sentiment analysis-based approach for the fake reviews classification and showed an average accuracy value of 96.30%.

The values shown in Table 2 are clearly depicting that our work is more accurate to classify real and bogus product reviews in comparison to the peer approaches. The FRD-LSTM framework acquired the average accuracy value of 97.21% along with the precision, recall, and F1-Score of 96%, 97.24%, and 96.61% respectively which are larger compared to the latest approaches mentioned in Table 2. More descriptively, the comparative approaches show an average accuracy value of 88% which is 97.21% for our approach. Hence, for the accuracy metric, we have given an average performance gain of 9.21%. Moreover, for the precision

Table 2 Comparison of the FRD-LSTM model with the latest methods

Method	Precision	Recall	F1-Score	Accuracy
Alsubari et al. [5]	87%	85%	86%	87%
Hajek et al. [13]	—	—	82.50%	89.30%
Kumar et al. [17]	73%	73.80%	73.30%	78.20%
Baishya et al. [7]	—	—	—	96.30%
Proposed (FRD-LSTM)	96%	97.24%	96.61%	97.21%

measurement, the peer methods show an average value of 80%, which is 96% for the presented technique. So, for the precision measure, we have attained an average performance gain of 16%. Furthermore, for the recall and F1-Score, the competitor approaches show the average value of 79%, and 81% respectively, which are 97.24% and 96.61% for our work. Therefore, we have presented the average performance gains of 18%, and 16% for the recall and F1-Score measures respectively. The main reason for the better fake reviews classification performance of our work is that the comparative techniques use a very complex network architecture, which eventually results in a model overfitting problem. While in comparison, we have used the DCWR model with the PCA algorithm which presents a more nominative set of text features while reducing the feature space as well. Here, the inclusion of the PCA approach optimizes the feature selection power of the proposed approach. Such structure of the proposed model prevents it from encountering the model overfitting problem. Moreover, the inclusion of the Bi-LSTM classifier assists in better tackling complicated text transformations and is much less susceptible to the vanishing gradient problem. Furthermore, the Bi-LSTM classifier is much better at handling long-term dependencies due to its ability to remember information for extended periods of time. Therefore, we can say that we obtained better classification performance due to the usage of deep features with the Bi-LSTM approach that responsibly represents the complicated text transformations due to its high recall ability which in turn enhances the fake reviews classification results of our approach. Additionally, the proposed approach may lack to tackle the reviews based on sarcasm.

5 Conclusion

Fake reviews posted online about the products can mislead buyers which can cause them to lose the trust of the audience in online shopping. To overcome the challenges, we have presented a DL-based approach for the accurate detection of real and fake reviews. Initially, a preprocessing step is applied to remove unnecessary details from the textual data. Then, the DCWR algorithm is used for calculating the features from the processed text. In the next phase, the PCA method is applied to the computed key points to reduce the feature vector size. Lastly, the Bi-LSTM classifier is trained on the reduced features to execute the reviews classification task as being real or fake. An extensive evaluation is executed on the Amazon product reviews dataset to show the efficacy of our approach. From the conducted analysis, we can say that the deep features extracted with the DCWR approach along with the Bi-LSTM classifier can lead to robust accuracy. Exceptionally, the introduced framework outperforms the existing latest fake review classification approaches in terms of accuracy, precision, recall, and F1-Score at 97.21%, 96%, 97.24%, and 96.61% respectively. Extensive experiments have been carried out as compared to other latest fake reviews classification approaches by using a standard database containing diverse and unbalanced data to demonstrate the accuracy of the introduced model. In the future, we plan to test our method over more challenging and real-time datasets. Moreover, we plan to consider the reviews based on sarcasm.

Data availability We have used standard dataset which is publicly available.

Declarations

Conflict of interest There is no conflict of interest between authors.

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